NaturArv3D Data Pipeline

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Introduction

- Compression study somewhat on hold
  - Until more details hashed out
  - Format requirements, data size expectations
- Very helpful input from SNM — *many thanks Mark, Christy, Thomas, Joni!*

Format Support

Format	ImageJ/Fiji	Napari	VGSTUDIO	Amira/Avizo	3DSlicer	Dragonfly	Checkpoint	Morphosource	Data Types	Comment
BioFormats	Yes[1]	Yes[1]	No	Yes	No	No	No	No		
TIFF	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	u/i[8/16/32], f16/32	
BigTiff	Yes	Yes[1]	No?	Yes	No?	No?	No?	No?	u/i[8/16/32], f16/32	
JPEG-1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	u8 (u16 LL)	
JPEG-2000	Yes	?	Yes	Yes	No?	No?	No?	Yes?	u1-38	
JPEG-XR	No[2]	?	No?	No[2]	No?	No?	No?	No?	u8/16/32	
JPEG-LS	No?	?	No?	No?	No?	No?	No?	No?	u8/16/32	
JPEG-XL	No	Yes[1]	No	No	No	No	No	No	u8/16/32	
DICOM	Yes	Yes[1]	Yes	Yes	Yes	Yes	Yes	Yes	u8/16	Likely no JXL support?
OME-XML	Yes[1]	Yes[1]	No	Yes	No	No	No	No	u/i[8/16/32], f16/32, …	
OME-TIFF	Yes[1]	Yes[1]	No	Yes	No?	No?	No?	No	u/i[8/16/32], f16/32, …	
OME-ZARR	Yes[1]	Yes[1]	No	Yes[3]	No	No	No	No	u/i[8/16/32], f16/32, …	
HDF5	Yes	Yes[1]	No[4]	Yes[1]	No	Yes?	No	No	u/i[8/16/32], f16/32, …	
PNG	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes?	u8/16	
PPM	Yes	?	Yes	Yes	No?	No?	No	No	u8/16	
RAW	Yes	?	Yes	Yes	Yes	Yes	No	No?		
BMP	Yes	?	Yes	Yes	Yes	Yes	No	Yes	u8	
Analyze/HDR	Yes?	Yes[1]	No	Yes	Yes	Yes	Yes	No	u8, i16/32, f32/64, …	
NifTI	Yes	Yes[1]	No	Yes	Yes	Yes	Yes	No	u/i[8/16/32], f32/64, …	
Other notable					Many		FITS (.fits), Kitware VTK (.vtk)			
Source			E-Mail with *Technical Solutions Export*		3D Slicer Wiki	Dragonfly Manual	Checkpoint	Duke		
Comment	FOSS (GPLv3)	FOSS (BSD-3), Plugin system	Proprietary	Proprietary	FOSS (BSD), Anything via Python		Proprietary	Museum data repository		

[1] With plugin
[2] Supported through `.czi` format
[3] With plugin by Janelia
[4] Does support HDF4

Software Support: Summary

- Not widespread support for formats modern formats
  - Wonder how good support would actually be for a 256GB TIFF stack in practice?
- Large datasets have been rare, but is changing
- DICOM format very promising
  - Supports JPEG-LS, JPEG-2000
  - Only widely supported format with lossy compression
  - Even supports JPEG-XL as of 2025
    - Likely implementors do not yet comply
    - Can beat implementors over the head if they state DICOM support
  - Have been very impressed with JPEG-XL
    - Mayo: 50-60% size reductions over existing lossy compressions with no visual difference[1]
- Opinion: ~~Insane~~ Silly to reduce bit-depth to 8-bit but not use (lossy) compression

[1] https://www.dicomstandard.org/news-dir/current/docs/sups/sup232-slides.pdf

Example: Czeibart, UHR Skulls

Kalman Czeibart, 430 skull CT scans (Christy)

- Source: https://nc.elte.hu/s/doZnwPjgeBaZ7as
- NifTI, gzip, `int16`, $\approx 500 \times 500 \times 1000$
- Sample `001_A_AIRT`: 142MB compressed, 234M uncompressed
  - $\mathrm{CR} = 1.65$, $\mathrm{SS} = 39\%$
- 430 samples = 47.9GB compressed
- Wide software support

If we have similar size, no need to bother with lossy or non-standard compression!

Examples: Beamlines

- EMBL/DESY P14 beamline: TIFF/OME-Zarr/WEBKNOSSOS, see here
- MAX IV: HDF5
- Most beamlines store data uncompressed/losslessy compressed
  - Those that have switched to Zarr likely use Blosc (lossless)

Even beamlines with huge data storage requirements have not yet adopted lossy compression widely.

But note: they have switched to chunked storage formats, likely out of necessity!

MorphoSource

- Samples in their CT collections seem to be mostly in 50MB – 2GB range
- Supports formats: TIFF, JPEG-1, BMP, DICOM[1]
  - Not even NifTI support
- Institutions with deposits over 1TB expected to pay
  - Fee is 3.47 USD/GB + 30 USD flat fee[2]
- **Question**: *Is this the intended/expected/desired storage solution for NaturArv3D?*

[1] https://duke.atlassian.net/wiki/spaces/MD/pages/35422892/Critical+Formatting+Details
[2] https://duke.atlassian.net/wiki/spaces/MD/pages/274268722/Data+Deposit+Costs

Size Considerations

- Detectors:
  - Large samples: Spectrumlogic, 5606×4800 px, pixel size $50\,\mu\text{m}$
  - Smaller samples: Varex, 3072×3072 px, pixel size $139\,\mu\text{m}$
- *Optimistic* canonical volume: 3072^3 vx $= 28.9$ Gvx (assuming 16-bit)
- Assume 50% (?) crop as an average: 14.4 Gvx
  - 16-bit: $28.8\,\text{GB}$ uncompressed

Cloud storage cost: Backblaze B2 Object Storage[1]:

- Includes 300% monthly egress (data can be downloaded 3 times per month for before additional fees)
- Price is for 5 years of redundant, 'active archive' storage
- 83 USD / TB / year $\Rightarrow$ 0.415 USD / GB $(2.65$ DKK$)$
  - Significantly cheaper than MorphoSource, though this is only for 5 years

[1] https://www.backblaze.com/cloud-storage/pricing, accessed 2026-08-26

Size Consideration Table

Data / Scheme	*Czeibart* 500×500×1000 0% crop		16-bit	2^3 × *Czeibart* 1000×1000×2000 0% crop		16-bit	***Small Detector*** 3072×3072×3072 50% crop		16-bit	*Large Detector* 5606×5606×4800 50% crop		16-bit
Uncompressed	500 MB	250 GB	13.0 TB	4.00 GB	2.00 TB	104 TB	29.0 GB	14.5 TB	756 TB	151 GB	75.4 TB	3.94 PB
CR = 1	11.1 kr	5.54 kkr	289 kkr	88.7 kr	44.3 kkr	2.31 Mkr	643 kr	321 kkr	16.8 Mkr	3.34 kkr	1.67 Mkr	87.3 Mkr
SS = 0.00%	1.33 kr	663 kr	34.6 kkr	10.6 kr	5.30 kkr	277 kkr	76.9 kr	38.4 kkr	2.01 Mkr	400 kr	200 kkr	10.4 Mkr
Lossless	294 MB	147 GB	7.67 TB	2.35 GB	1.18 TB	61.4 TB	17.1 GB	8.53 TB	445 TB	88.7 GB	44.4 TB	2.32 PB
CR = 1.7	6.52 kr	3.26 kkr	170 kkr	52.2 kr	26.1 kkr	1.36 Mkr	378 kr	189 kkr	9.87 Mkr	1.97 kkr	984 kkr	51.3 Mkr
SS = 41.17%	780 mkr	390 kr	20.3 kkr	6.24 kr	3.12 kkr	163 kkr	45.2 kr	22.6 kkr	1.18 Mkr	235 kr	118 kkr	6.14 Mkr
Lossy	20.0 MB	10.0 GB	522 GB	160 MB	80.0 GB	4.17 TB	1.16 GB	580 GB	30.3 TB	6.03 GB	3.02 TB	157 TB
CR = 25	443 mkr	222 kr	11.6 kkr	3.55 kr	1.77 kkr	92.6 kkr	25.7 kr	12.9 kkr	671 kkr	134 kr	66.9 kkr	3.49 Mkr
SS = 96.00%	53.0 mkr	26.5 kr	1.38 kkr	424 mkr	212 kr	11.1 kkr	3.08 kr	1.54 kkr	80.2 kkr	16.0 kr	8.00 kkr	417 kkr
Lossy	10.0 MB	5.00 GB	261 GB	80.0 MB	40.0 GB	2.09 TB	580 MB	290 GB	15.1 TB	3.02 GB	1.51 TB	78.7 TB
CR = 50	222 mkr	111 kr	5.78 kkr	1.77 kr	887 kr	46.3 kkr	12.9 kr	6.43 kkr	335 kkr	66.9 kr	33.4 kkr	1.75 Mkr
SS = 98.00%	26.5 mkr	13.3 kr	692 kr	212 mkr	106 kr	5.53 kkr	1.54 kr	769 kr	40.1 kkr	8.00 kr	4.00 kkr	209 kkr
Lossy	5.00 MB	2.50 GB	130 GB	40.0 MB	20.0 GB	1.04 TB	290 MB	145 GB	7.56 TB	1.51 GB	754 GB	39.4 TB
CR = 100	111 mkr	55.4 kr	2.89 kkr	887 mkr	443 kr	23.1 kkr	6.43 kr	3.21 kkr	168 kkr	33.4 kr	16.7 kkr	873 kkr
SS = 99.00%	13.3 mkr	6.63 kr	346 kr	106 mkr	53.0 kr	2.77 kkr	769 mkr	384 kr	20.1 kkr	4.00 kr	2.00 kkr	104 kkr

- Cell rows: data size, storage cost (MorphSource), storage cost (Backblaze B2)
- Cell columns: 1 recon., 1 week of recons. (shaded, 500), 1 year of recons. (26089)
- Units have been SI-ified: mkr = milli-kroner, Mkr = mega-kroner/million, etc.

Potential Storage Solutions

- Lowest Common Denominator
  - Appears to be uncompressed TIFF/DICOM/JPEG-1/PNG
  - Note: Likely that many of them don’t support JPEG-XL in DICOM yet?
- Zarr + Objects (+ Local Conversion)
  - Build or find small tool that can convert Zarr from object storage to TIFF locally
- A Reasonably Common Denominator
  - HDF5 (like MAX IV and others, likely still spotty codec support)
- Zarr/OME-Zarr/TIFF/… + REST API
- …?

The Case for Zarr

- Zarr has seen rapid adoption
  - NASA is commited — VEDA/ODD team
  - OME-Zarr is leading NGFF (Next Generation File Format) selected by OME
  - EMBL is adopter (DESY beamlines, Image Data Resource (IDR))
  - Janelia is adopter (OpenOrganelle, other projects)
  - + Many other prestigious and important institutions
- Zarr is vastly technologically superior to legacy formats
  - Access: Chunking, parallel access, multi-dimensional, cloud friendly (object storage)
  - Efficiency: **Codecs**, streaming/async, performant
  - Open: **Open standard, open governance**, highly interoperable
- Opinion: **Zarr/OME-Zarr is the correct long-term solution**
  - *May not be acceptable here due to legacy software support and data repository requirements*

Extra: JPEG-XL Lossy from Last Meeting

- 3D reconstructions take up bulk of storage
- Can be remade from archived projections
- This enables lossy compression
- Lossy JPEG-XL:
  - Distance 1 (middle row) gives 98.9% space saving
    - CR=90.5, SSIM=0.9992
  - Distances >1 produce noticeable artefacts and smoothening
  - Distance 0.1 gives 92.1% space saving
    - CR=12.7, SSIM=0.9995
- Demonstrates how good lossy compression is
  - mostly for SNM's benefit

